Available online at www.sciencedirect.com
ScienceDirect
Neurotoxicology and Teratology 29 (2007) 219–227
NEUROTOXICOLOGY AND TERATOLOGY
www.elsevier.com/locate/neutera

Oral Accutane® (13-*cis*-retinoic acid) has no effects on spatial learning and memory in male and female Sprague-Dawley rats

Sherry A. Ferguson *, Kimberly J. Berry

Division of Neurotoxicology, National Center for Toxicological Research/FDA, 390 NCTR Road, Jefferson, AR 72079, United States

Received 6 September 2006; received in revised form 13 October 2006; accepted 24 October 2006
Available online 1 November 2006

Abstract

Descriptions of psychiatric effects with Accutane® (13-*cis*-retinoic acid (13-*cis*-RA)) use prompted a series of studies in a rodent model to ascertain its cognitive effects. Previously, we reported no effects on measures of anhedonia and depression in rats treated with 7.5, 22.5, or 30 mg/kg 13-*cis*-RA [S.A. Ferguson, F.J. Cisneros, B. Gough, J.P. Hanig, K.J. Berry, Chronic oral treatment with 13-*cis*-retinoic acid (isotretinoin) or all-trans-retinoic acid does not alter depression-like behaviors in rats, Toxicol. Sci. 87 (2005) 451–459 [16]; S.A. Ferguson, F.J., Cisneros, J.P. Hanig, K.J. Berry, Chronic oral treatment with Accutane (13-*cis*-retinoic acid) does not increase measures of anhedonia or depression in male and female Sprague-Dawley rats, (in preparation) [19]]. Here, we assessed spatial learning and memory in male and female Sprague-Dawley rats gavaged daily beginning on postnatal day (PND) 59 with vehicle control (soybean oil), 7.5 or 30 mg/kg of 13-*cis*-RA. We have reported that 7.5 mg/kg produces serum levels of 13-*cis*-RA comparable to those of humans prescribed Accutane® [S.A. Ferguson, P.H. Siitonen, F.J. Cisneros, B. Gough, J.F. Young, Steady state pharmacokinetics of oral treatment with 13-*cis*-retinoic acid or all- trans-retinoic acid in male and female adult rats, Basic Clin. Pharmacol. Toxicol. 98 (2006) 582–587 [18]]. Three behavioral tasks assessed spatial learning and memory after chronic 13-*cis*-RA treatment: the escape-reinforced Morris water maze (PNDs 111–115), the food-reinforced 8-arm radial maze (PNDs 132–136), and the water-reinforced NCTR complex maze (PNDs 153–157). Behaviors were measured after a minimum of 52 and maximum of 94 days of 13-*cis*-RA treatment. 13-*cis*-RA treatment had no effects on performance of the 8-arm radial maze or the NCTR complex maze. Treatment effects on Morris water maze performance were negligible and neither dose-related nor consistent. Performances of the control group were quite similar to those previously described in this laboratory. These results indicate that chronic 13-*cis*-RA treatment in male and female rats has few effects on measures of spatial learning and memory.
© 2006 Elsevier Inc. All rights reserved.

Keywords: Accutane; Isotretinoin; Learning; Memory; Water maze; Radial maze

1. Introduction

Severe recalcitrant nodular acne responds extremely well to oral treatment with Accutane® (isotretinoin, 13-*cis*-retinoic acid (13-*cis*-RA)) and as a result, use of 13-*cis*-RA continues to increase [37,39]. Relative to what is known about the functions or effects of exogenous and endogenous retinoic acids (RAs) during development, very little is known about the role of RAs in the adult brain. However, accumulating evidence strongly suggests that RAs have direct and indirect roles in the normal functioning of the adult central nervous system [28,29].

Animal studies have implicated RAs in processes involved in learning and memory (reviewed in [28,29]). Much of this work has examined the effect of RA deficiency on the hippocampus or its functioning via behavioral tasks. For example, knockout mice lacking a specific RA receptor (i.e., RARβ) exhibited poor Morris water maze performance and impaired hippocampal long-term potentiation and depression [5]. Mice lacking both the RARβ and RXRγ receptors were also deficient in tasks measuring spatial memory [38]. Dietary vitamin A deprivation in mice resulted in 8-arm radial maze deficits [12] as well as deficits in hippocampal long-term potentiation and depression which were then reversed with all-*trans*-retinoic acid treatment

* Corresponding author. HFT-132, 3900 NCTR Road, Jefferson, AR 72079, United States. Tel.: +1 870 543 7589; fax: +1 870 543 7181.
E-mail address: Sherry.Ferguson@fda.hhs.gov (S.A. Ferguson).

0892-0362/$ - see front matter © 2006 Elsevier Inc. All rights reserved.
doi:10.1016/j.ntt.2006.10.008

[31]. Similarly, vitamin A deprivation in rats resulted in 8-arm radial maze deficits, smaller nuclei of hippocampal CA1 neurons, and decreased hippocampal acetylcholine release [7].

The effects of exogenous treatment with excess RAs have been less studied in adults. Two reports described increased errors in the 8-arm radial maze in adult rats and mice treated with 13-*cis*-RA [8,10]. Specifically, mice treated intraperitoneally with 13-*cis*-RA for 28 days exhibited increased errors (defined as re-visiting a previously visited arm or not consuming the food reinforcement in an arm) in the 8-arm radial maze over 5 daily trials [8]. Similarly, male rats treated orally with 13-*cis*-RA were somewhat impaired in a two-stage discrimination task during treatment but exhibited slightly better performance during a follow-up test after RA treatment ended [10].

Our previous work has shown that oral (gavage) treatment with 7.5 mg/kg 13-*cis*-RA in male and female rats causes little gross toxicity and results in peak serum levels of ≈577 ng/ml [6,17,18,25]. Those serum levels are comparable to the 300–800 ng/ml achieved by humans undergoing standard 13-*cis*-RA treatment for acne [27,4,1,32,35,22]. In the current study, oral treatment of 7.5 or 30 mg/kg 13-*cis*-RA continued daily for 19 weeks during which three different measures of learning and memory were conducted: the Morris water maze, the 8-arm radial maze, and a dry land maze (the NCTR complex maze). Each of these spatial tasks employed a different type of reinforcer (i.e., water escape, food reinforcement, or water reinforcement).

2. Materials and methods

2.1. Animals

Complete descriptions of animals and treatment are available [19] and will only be summarized here. Postnatal day (PND) 21 Sprague-Dawley rats (45 male and 45 female, divided into 3 treatment groups of 15 rats/sex/group) were obtained from the National Center for Toxicological Research/FDA breeding colony and individually housed in standard polycarbonate tub cages lined with wood chip bedding. Ad libitum pelleted rat chow (NIH-31, Purina Mills, St. Louis, MO) and water were provided. All animal procedures were approved by the NCTR Institutional Animal Care and Use Committee.

2.2. 13-cis-retinoic acid treatment

Capsules (40 mg each) of 13-*cis*-RA (Accutane®, Roche Pharmaceuticals, Nutley, NJ) were obtained from Washington Wholesale Drug Exchange (Savage, MD). Each capsule was opened and its contents diluted in soybean oil (ICN Biomedicals, Irvine, CA) to produce 7.5 and 30 mg/kg suspensions. Soybean oil was used as this is one of the major constituents of Accutane® gelatin capsules [23]. Suspensions were prepared every two days and maintained at room temperature in amber glass containers to prevent photoisomerization. All lab areas and housing rooms were fitted with UV filters over fluorescent lights.

Treatment group assignment was based on PND 41 body weight such that there were approximately equal body weights in each group. On PNDs 57–58, each rat was gavaged with soybean oil to facilitate adaptation to the gavage procedure. Beginning on PND 59, each rat was gavaged daily with 0 (soybean oil only), 7.5 (low) or 30 (high) mg/kg of 13-*cis*-RA (n = 15/sex/dose). Gavage volume was 1.43 ml/kg and all gavages were done in the morning hours and always prior to any behavioral testing.

2.3. Behavioral testing

Three behavioral tests were conducted in this study: the Morris water maze, the 8-arm radial maze, and the NCTR complex maze. During the course of the study, the subjects underwent additional behavioral testing which has been [19] or will be reported separately. These tests were: open field activity testing (PNDs 51, 129), saccharin solution intake (PNDs 52–55, 73–76, 94–97, 115–118), unrewarded NCTR complex maze activity (PND 66), forced swim test (PNDs 83–84, 141–142), residential running wheel activity (PNDs 85–92), elevated plus maze testing (PND 98), residential figure 8 maze activity (PNDs 99–106), and emergence (light/dark box) testing (PND 125).

2.4. Morris water maze

On PNDs 111–115 (i.e., after 52 days of 13-*cis*-RA treatment), spatial learning and memory were assessed as previously described [14]. The water maze was located in a diffusely lit testing room and consisted of a circular stainless steel tank (183 cm interior diameter) with a black interior filled to a depth of 34 cm with 25–27 °C water which was made opaque by the addition of powdered black paint. An escape platform (10 cm in diameter) was made of Plexiglas and covered with a coarse material that provided grip for climbing which was located approximately 1 cm below the water surface. Except for probe trials, the platform remained in the same location for the 5 test days. Each rat was tested for 3 acquisition trials/day for each of 5 days, with one of 4 starting locations varied between trials. Testing order varied across days. If the platform was not located within the maximum time allowed for each trial (120 s), the rat was gently guided to the location. Once on the platform, the rat was allowed 20 s for spatial orientation before being removed to a holding cage. Intertrial intervals ranged 15–140 s and between each trial, the water was gently stirred to eliminate the potential of olfactory cues from previous swim patterns. A single 30 s probe trial in which the platform was removed and all subjects started from the same location was conducted at the end of test days 3 and 5. Probe trials conducted in the middle of 5 days of training trials do not disrupt learning and such trials conducted immediately after training trials represent an assessment of working memory [2]. Probe trials are typically 60 s in length; however, because such a duration may be more sensitive to persistence than spatial preference [21], the duration here was shortened to 30 s. For each acquisition trial (i.e., the 3 daily trials on test days 1–5), latency to find the platform, path length (cm), swim speed (m/s), percentage of time engaged in slow

swimming (defined as <5 cm/s), and percentage of thigmotaxis time (defined as swimming in 10% of the area closest to the edge of the maze) were recorded by a video-tracking/computer digitizing system (HVS Image, Hampton, UK). For rats not finding the platform, the maximum latency of 120 s was recorded. From these measures, average and cumulative proximity measures for each trial were calculated by the HVS Image program. The average proximity measure is similar to that described by Gallagher et al. [20] and is determined by sampling the position of the rat in the water maze at a fast rate (10×/s) to provide a record of its average distance from the platform during each trial. The cumulative proximity measure is similar to the average proximity measure but is instead the sum of the distances from the platform on each trial. For each of the two probe trials, path length (cm), swim speed, percentage of time engaged in slow swimming, and percentage of time in the quadrant in which the platform was previously located were recorded. Acquisition and probe trials were analyzed separately. Prior to statistical analyses, the three daily acquisition trials were averaged to provide a single value for each endpoint for each day. Because latency to find the platform and path length are highly correlated (here, $r = 0.95$ for all rats, all sessions), only latency was analyzed.

2.5. 8-Arm radial maze

On PNDs 132–136 (i.e., after 73 days of 13-*cis*-RA treatment), spatial learning and memory were assessed in the 8-arm radial maze (Coulbourn Instruments, Allentown, PA). The apparatus consisted of a central octagonal arena (26 × 24.5 cm) with eight arms radiating outward. Each arm was 35 × 8.5 × 11.5 cm with a doorway opening from the central arena of 7.5 × 6.5 cm into the arm. At the end of each arm was a smaller opening (3.5 × 4.5 cm) in which a food pellet could be placed. Beginning on PND 130, each rat was restricted to 1 h ad lib access to food which was provided after the daily gavage and behavioral testing. On PND 132, one 5 min habituation session was conducted in which the rat was acclimated to the noise of the door(s) openings and closings and during which time, it could consume 12 reinforcement pellets (45 mg dustless precision chocolate pellets, Bioserve, Frenchtown, NJ) which were placed in the central arena. On PNDs 133–136, the end of each the 8 arms was baited with a single reinforcement pellet and the rat was placed in the central arena. After 10 s, all 8 doors opened and the latency to enter all 8 arms and consume the pellets was recorded. If the rat did not consume all 8 pellets within 5 min, the session ended. For each of the 4 days (PNDs 133–136), the total number of arms entered, number of

Fig. 1. The NCTR complex maze. Dimensions are in cm. The right and left wire grid tops were lowered after placing the rat into the maze and raised to remove the rat from the maze. Each rat was tested on one of two maze programs. Maze A required the rat to shuttle between goal arms A_1 and A_2. Maze B required the rat to shuttle between goal arms B_1 and B_2. Although water dispensers are shown on arms other than the four goal arms, these were not used. Spatial cues (e.g., posters on the walls) are not shown. (Photograph and editing by Richard Rasmussen and Jeff Runnells.)

reinforcers obtained, number of entries into arms before revisiting a previously visited arm (entries to repeat (ETR)), and average duration in each arm (total test time/total number of arms entered) were recorded.

2.6. NCTR complex maze

On PNDs 153–157 (i.e., after 94 days of treatment), activity and learning performance were assessed for 5 consecutive days similar to that previously described [24]. The apparatus was an array of 24 acrylic arms (24.3 × 12.5 × 12.0 cm) open at the top and covered with a large wire mesh, hinged at each side to allow easy access (see Fig. 1). A photobeam was located at the center of each arm and beam breaks were monitored via a computer interface. Two of the arms were consistently designated as goal arms (1–2 drops of tap water per reinforcer). The room was illuminated by diffuse red lighting. The paradigm required the rat, which had been water-deprived for 24 h, to shuttle between two goal arms (Goal Arms A_1 and A_2 or B_1 and B_2 in Fig. 1) to obtain reinforcers. There was no limit on the number of reinforcers a rat could obtain in any given 15 min session. Each rat was allowed 30 min of ad lib water immediately after each session and amount consumed during this time was measured daily. Variables assessed for each session included total activity (number of arm entries), total reinforcers earned, and an efficiency ratio (100 * (number of reinforcers earned/number of arm entries)).

2.7. Statistical analyses

Data from each of the three learning tasks were analyzed via repeated measures analysis of variance (ANOVA) with treatment, sex, and day (or session) as factors. Post-hoc comparisons of significant main effects or interactions were conducted using Tukey's HSD with $p < 0.05$ required for significance.

3. Results

Results of 13-*cis*-RA treatment on body weight, food and water intake of these subjects have been reported [19]. In brief, 13-*cis*-RA treatment had no effects on body weight or food intake. There was a moderate effect of 13-*cis*-RA treatment to decrease water intake, particularly in females.

3.1. Morris water maze

3.1.1. Acquisition trials

Analysis of latency to find the platform indicated a significant interaction of treatment × sex × day ($F(8, 336) = 2.73$, $p < 0.007$); however, there were no significant post-hoc tests on any specific test day between males and females within each treatment group (i.e., no differences between control males and females, between 7.5 mg/kg males and females or between 30 mg/kg males and females) or between same-sex treatment groups on any specific test day (i.e., no differences between control females and 7.5 or 30 mg/kg females and no differences between control males and 7.5 or 30 mg/kg males). A significant main effect of day ($F(4, 336) = 127.66$, $p < 0.001$) indicated that latencies decreased over the five testing days. The two way interactions (treatment × sex, sex × day, and treatment × day) as well as the main effects for treatment and sex were not statistically significant in the analysis of latency to find the platform. Fig. 2 shows latency to find the platform for the five test days (shown separately by treatment group and sex for clarity). The ANOVA of percentage of time in slow swimming indicated a significant interaction of treatment × sex × day ($F(8, 336) = 2.05$, $p < .05$); however, post-hoc tests indicated no significant differences on any test day between males and females within each treatment group or between same-sex treatment groups. A significant main effect of sex ($F(1, 84) = 15.43$, $p < .0002$) indicated more slow swimming in females. The two way interactions and the main effects for treatment, sex, and day were not statistically significant. Analysis of percentage of time engaged in thigmotaxis indicated a significant interaction of sex × day ($F(4, 336) = 8.00$, $p < 0.001$) and post-hoc tests indicated that while females engaged in more thigmotaxis than males on test days 2–5, this difference was significant on test days 4 and 5 ($p < 0.05$). Both sexes exhibited more thigmotaxis on the first test day relative to test days 2–5 ($p < .05$). Significant main effects of sex

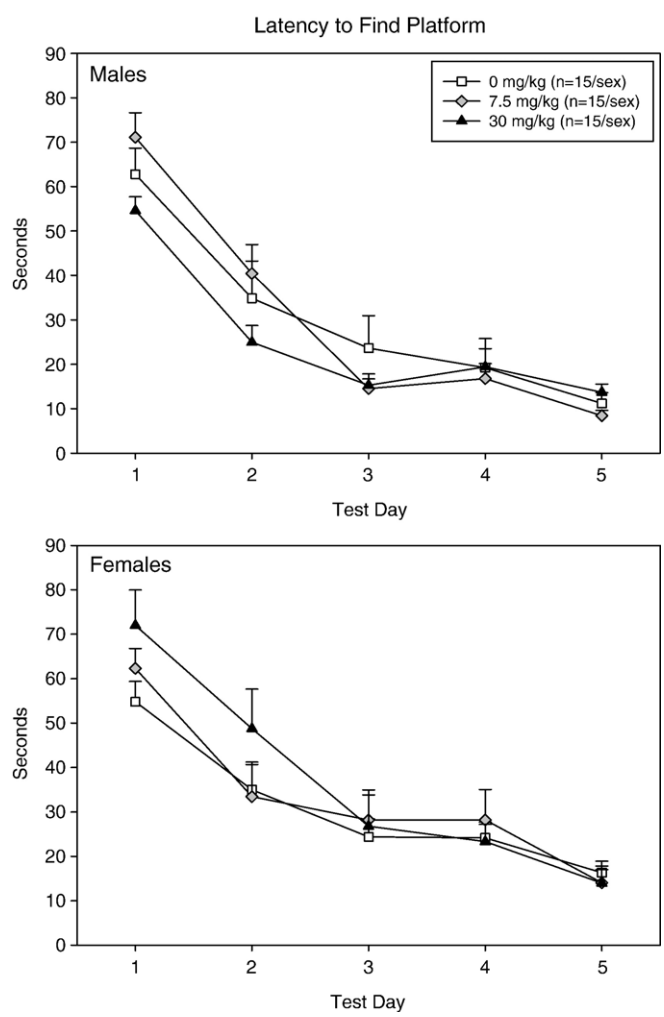

Fig. 2. No effects of 13-*cis*-RA treatment on Morris water maze performance. Latency to find platform (top: males; bottom: females).

($F(1, 84)=8.11$, $p<006$) and day ($F(4, 336)=89.74$, $p<.0001$) revealed that females engaged in more thigmotaxis and that thigmotaxis was most prevalent on the first day. The three way interaction, the remaining two way interactions as well as the main effect for treatment were not statistically significant.

Analysis of the average proximity measure indicated a significant interaction of treatment × sex × day ($F(8, 336)=1.97$, $p<.05$); however, post-hoc tests indicated no significant differences on any test day between males and females within each treatment group or between same-sex treatment groups. A significant main effect of day ($F(4, 336)=86.02$, $p<.0001$) indicated decreasing average proximity across days. The two way interactions and the remaining main effects of treatment and sex were not statistically significant. Analysis of the cumulative proximity measure indicated a significant interaction of treatment × sex × day ($F(8, 336)=2.69$, $p<.008$); again, however, post-hoc tests indicated no significant differences on any test day between males and females within each treatment group or between same-sex treatment groups. A significant main effect of day ($F(4, 336)=148.13$, $p<.0001$) indicated decreasing cumulative proximity across days. The two way interactions and the remaining main effects of treatment and sex were not statistically significant. Swim speed analysis indicated two significant interactions: treatment × sex ($F(2, 84)=3.11$, $p<.05$) and sex × day ($F(4, 336)=6.58$, $p<.0001$). Post-hoc tests indicated significant sex differences in the two 13-*cis*-RA groups, but not in the control group. Specifically, the 7.5 and 30 mg/kg 13-*cis*-RA treated females swam slower than their same treatment male counterparts ($p<.05$); however, while control females swam slower than control males on test days 2–5, this difference was not significant (swim speed (cm/s) for 0 mg/kg males and females = 25.2 ± 0.8 and 24.0 ± 1.0, respectively; 7.5 mg/kg males and females = 25.8 ± 0.7 and 22.2 ± 0.8, respectively; 30 mg/kg males and females = 26.6 ± 0.7 and 21.4 ± 0.7, respectively). Post-hoc tests of the sex × day interaction indicated that on all test days except the first test day, females swam slower than males ($p<.05$). Further, males on the first test day swam slower than they did on test days 2–5 ($p<.05$). Significant main effects of sex (1, 84) = 26.50, $p<.0001$) and day ($F(4, 336)=3.96$, $p<004$) indicated slower swimming in females and on the first day. The three way interaction, the remaining two way interaction (treatment × day) and the main effect of treatment were not statistically significant.

3.1.2. Probe trials

Analyses of path length and swim speed during each of the two 30 s probe trials both indicated significant interactions of sex × trial: ($F(1, 82)=5.85$, $p<.02$) and $F(1, 82)=5.88$, $p<.02$), respectively. Post-hoc tests indicated identical results in that on both trials, females had a shorter path length and slower swim speed than males ($p<.05$). Further, males on trial 2 (which occurred at the end of test day 5) had a shorter path length and a slower swim speed than on trial 1 (which occurred at the end of test day 3) ($p<.05$). Significant main effects of sex ($F(1, 84)=20.57$, $p<.0001$) and ($F(1, 84)=20.58$, $p<.0001$) for path length and swim speed, respectively, and probe trial ($F(1, 82)=21.33$, $p<.0001$) and ($F(1, 82)=21.29$, $p<.0001$) for path length and swim speed, respectively, indicated shorter path lengths and slower swimming in females as well as longer path length and faster swimming on probe trial 1. The three way interaction, remaining two way interactions and the main effect of treatment were not statistically significant in the analyses of path length and swim speed. Analysis of percentage of time engaged in slow swimming indicated significant main effects of treatment ($F(2, 84)=4.00$, $p<.03$) and trial ($F(1, 82)=6.79$, $p<.02$). Post-hoc tests indicated that the 30 mg/kg 13-*cis*-RA group exhibited a higher percentage of time engaged in slow swimming relative to the 0 mg/kg group ($p<.05$) (mean percentage ± SEM for 0, 7.5, and 30 mg/kg groups = 13.7 ± 1.4, 17.3 ± 2.4, and 20.0 ± 2.0, respectively). Further, all rats engaged in a higher percentage of time in slow swimming on trial 2 relative to trial 1 ($p<.05$). The three way interaction, two way interactions and the main effect of sex were not statistically significant. Finally, analysis of percentage of time spent in the correct quadrant indicated a significant interaction of sex × treatment ($F(2, 84)=3.57$, $p<.04$); however, there were no significant post-hoc tests (means ± SEM for 0, 7.5, 30 mg/kg males = 31.9 ± 1.8, 26.3 ± 2.3, and 37.4 ± 3.7, respectively) (means ± SEM for 0, 7.5, 30 mg/kg females = 26.7 ± 2.7, 30.7 ± 4.1, and 26.38 ± 2.8, respectively). The three way interaction, the remaining two way interactions as well as the main effects of treatment, sex and trial were not statistically significant.

3.2. 8-Arm radial maze

Analysis of total number of arms entered/session indicated significant main effects of sex ($F(1, 84)=34.05$, $p<.0001$) and session ($F(3, 252)=101.22$, $p<.0001$). Females entered more arms than males (9.26 ± 0.49 and 5.65 ± 0.38, respectively) and the number of arms entered on session 1 for all rats was significantly more than that entered on sessions 2–4 (number of arms entered for sessions 1–4 = 11.09 ± 0.47, 6.47 ± 0.40, 6.23 ± 0.41, 6.02 ± 0.40, respectively). The three way interaction, the two way interactions and the main effect of treatment were not statistically significant. Analysis of number of entries to repeat (ETR) indicated a significant interaction of sex × session ($F(3, 252)=6.40$, $p<.0003$). Post-hoc tests revealed that while females entered more arms before re-visiting an arm on sessions 2–4 than did males on the same sessions, this was significant only for sessions 3 and 4 ($p<.05$). In contrast, males entered fewer arms before re-visiting an arm on sessions 2–4 than they did on session 1 ($p<.05$). The three way interaction, the remaining two way interactions, and the main effect of treatment were not statistically significant in the analysis of ETR. Analysis of number of reinforcers obtained indicated significant main effects of sex ($F(1, 84)=24.62$, $p<.0001$) and session ($F(3, 252)=2.89$, $p<.04$). Post-hoc tests indicated that females obtained more reinforcers than males (5.72 ± 0.37 and 3.16 ± 0.37, respectively) and that more reinforcers were earned on session 4 than on session 1 ($p<.05$). Although it appears that the 7.5 and 30 mg/kg females earned somewhat fewer reinforcers than the 0 mg/kg females, the three way and two way interactions as well as the main effect of treatment were not

statistically significant. Fig. 3 shows number of reinforcers obtained for each sex and treatment group for the 4 test days. Analysis of average duration in each arm indicated a significant interaction of sex × session ($F(3, 238) = 3.90$, $p < .01$). Post-hoc tests indicated that females had a shorter average duration on session 1 than on session 4 ($p < .05$) and a shorter average duration on session 3 than did males on the same session ($p < .05$). Further, males had a shorter average duration on session 1 than on sessions 2–4 ($p < .05$). The three way interaction, the remaining two way interactions, and the main effect of treatment were not statistically significant.

3.3. NCTR complex maze

Analysis of number of reinforcers earned indicated a significant interaction of sex × session ($F(4, 272) = 5.56$, $p < .0003$). Post-hoc tests indicated that females earned more reinforcers during sessions 3–5 than during sessions 1 and 2 ($p < .05$). Females earned more reinforcers than males during sessions 1 and 3–5 ($p < .05$). Males earned more reinforcers during session 5 than during sessions 1 and 3 ($p < .05$). The three way interaction, the remaining two way interactions and the main effect of treatment were not statistically significant. Fig. 4 shows number of earned reinforcers by sex and session (shown separately by treatment group for increased clarity). Analysis of activity (total number of arm entries) indicated significant effects of sex ($F(1, 83) = 45.41$, $p < .0001$) and session ($F(4, 272) = 3.39$, $p < .01$). Females were more active than males ($p < .05$) (mean arm entries ± SEM: males = 169 ± 4, females = 226 ± 8) and all rats were more active on session 5 than on sessions 2 or 3 ($p < .05$) (mean activity ± SEM by session: session 1 = 197 ± 5, session 2 = 195 ± 7, session 3 = 193 ± 6, session 4 = 201 ± 6, session 5 = 207 ± 7). The three way and two way interactions as well as the main effect of treatment were not statistically significant. Analysis of the efficiency ratio indicated a significant interaction of sex × session ($F(4, 272) = 12.06$, $p < .009$). Post-hoc tests indicated that the efficiency ratios for females were significantly higher on sessions 3–5 than on session 1 or 2 ($p < .05$) and significantly

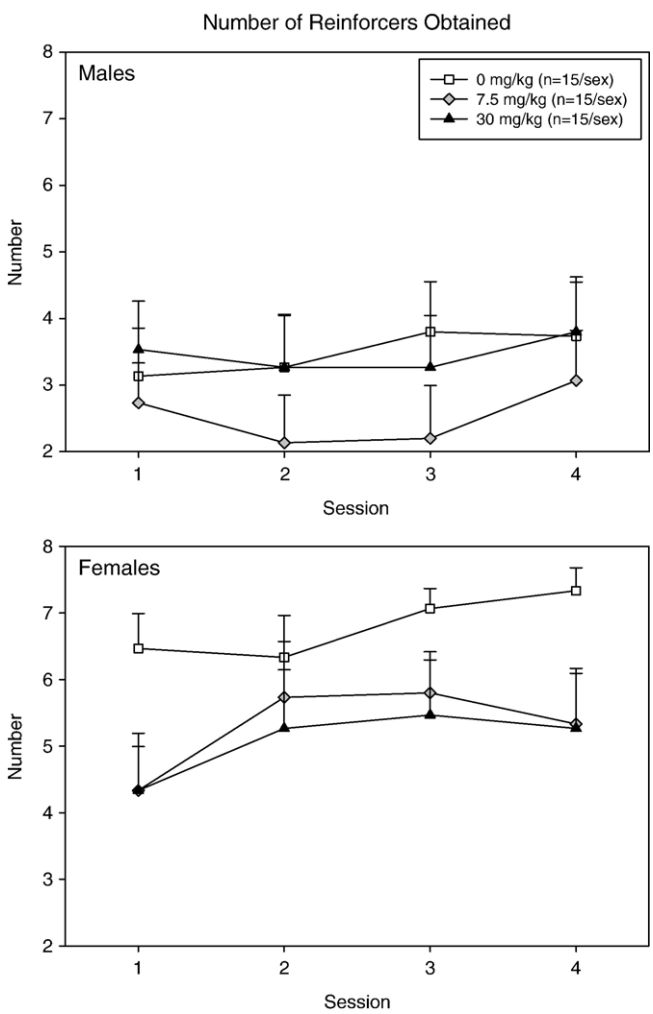

Fig. 3. Sex effects, but no effects of 13-*cis*-RA, on 8-arm radial maze performance. Number of reinforcers obtained (top: males; bottom: females). Note the larger number of reinforcers obtained by females.

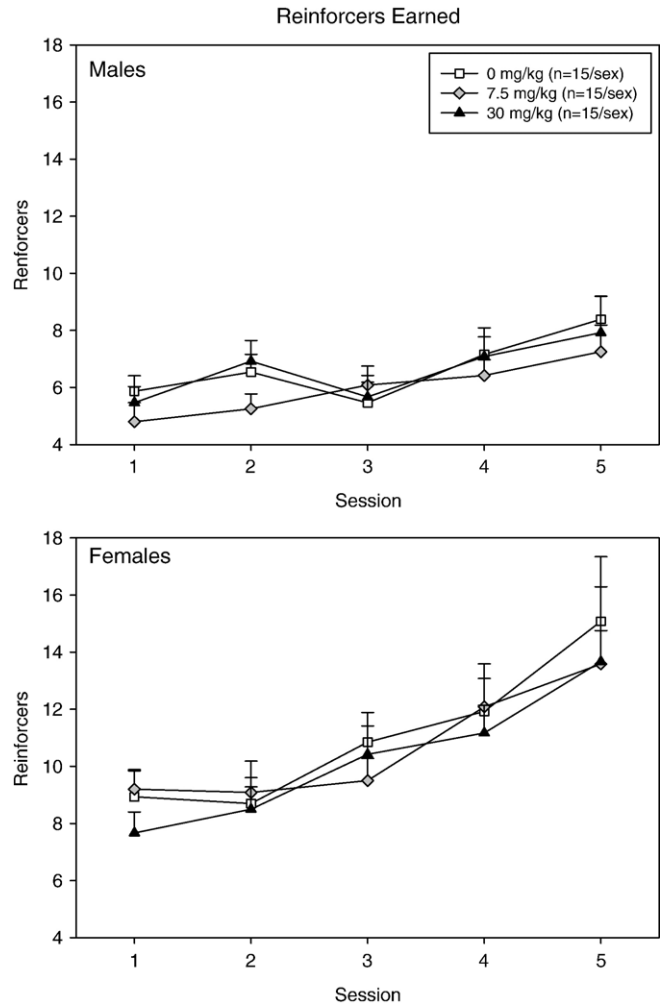

Fig. 4. No effects of 13-*cis*-RA treatment on number of reinforcers earned in the complex maze. Top: males. Males earned more reinforcers on session 5 than sessions 1 and 3 ($p < .05$). Bottom: females. Females earned more reinforcers on sessions 3–5 than sessions 1–2 ($p < .05$). Females also earned more reinforcers than males on sessions 1, 3, 4, and 5 ($p < .05$).

higher on session 5 than on session 3 ($p<.05$). Males had higher efficiency ratios on sessions 4 and 5 than on session 1 ($p<.05$) and a higher ratio on session 5 than on sessions 2–3 ($p<.05$). In addition, the efficiency ratios of females on sessions 3–5 were significantly higher than those of males on sessions 3–5 ($p<.05$). Significant main effects of sex ($F(1, 83)=16.81$, $p<.0001$) and session ($F(4, 272)=31.96$, $p<.0001$) indicated that females had higher efficiency ratios and that efficiency ratios increased over the 5 sessions. Amount of water consumed immediately after each test session did not differ with respect to treatment; however, females consumed more than males ($F(1, 83)=82.18$, $p<.0001$) (mean ml consumed/g body weight ± SEM: males = 0.0280 ± 0.0006, females = 0.0367 ± 0.0009). The three way and two way interactions as well as the main effects of treatment and session were not statistically significant in the analysis of amount of water consumed after each test session.

4. Discussion

Male and female Sprague-Dawley rats were treated with oral doses of 13-*cis*-RA which produced serum levels similar to those of human taking Accutane® and assessed on three measures of spatial learning and memory. 13-*cis*-RA treatment had no effects on performance of the food-reinforced 8-arm radial maze or the water-reinforced NCTR complex maze. Treatment effects on escape-reinforced Morris water maze performance were negligible and neither dose-related nor consistent. Data from the soy-oil treated control group were similar to those of previous studies in this laboratory indicating testing consistency and no quantifiable effects of gavage stress on these measures. Further, sex differences were similar to those previously reported. These results indicate that at doses which result in serum levels comparable to those of humans and that do not alter body weight or food intake, chronic 13-*cis*-RA treatment has few effects on spatial learning and memory in male and female rats.

Because of its particular sensitivity to hippocampal impairments (reviewed in [9]), the Morris water maze was predicted to be sensitive to 13-*cis*-RA treatment since such treatment has been shown to inhibit hippocampal neurogenesis [8,36]. However, rats treated with 7.5 or 30 mg/kg 13-*cis*-RA were as proficient as controls at locating the hidden platform. In no measure of acquisition (i.e., latency to find the platform, proximity endpoints) were there differences between the control group and either of the 13-*cis*-RA treated groups. Generally, probe trials are conducted on separate days from the days of training trials; however, as performed here on the same day, they represent a measure of working memory [2]. This working memory measure indicated no differences between treatment groups with the sole exception that the 30 mg/kg group engaged in more slow swimming than did the control group; however, this was not reflected in an overall slower speed. No group was impaired as indicated by percentage of time spent in the correct quadrant.

In general, female rats swam slower than males and exhibited increased thigmotaxis in the water maze during acquisition and probe trials. These are typical findings from many of our previous studies; however, they have not always reached statistical significance. Others have noted similar sex differences [33,3]. While males are often reported to have a significant advantage in the Morris water maze, this sex difference is strongest in paradigms that, unlike the current study, include pretraining trials [26].

Sex differences were more strongly apparent in results of the 8-arm radial maze. Nonetheless, 13-*cis*-RA treatment did not affect performance. The 8-arm radial maze methodology used here assessed working, not reference, memory as all arms contained reinforcers. Males are often reported to perform better in 8-arm radial maze tasks; however, a meta-analysis review of sex differences described smaller and less reliable sex differences in working memory only tests, particularly when animals were reared in isolation [26] as was done here.

Previous descriptions of radial maze deficits in animals treated with 13-*cis*-RA are difficult to reconcile with the current results. Crandall et al. [8] described deficits in mice treated intraperitoneally with 13-*cis*-RA using a similar radial maze methodology; however, species or administration route differences may explain the differing results. Dopheide and Morgan [10] treated male rats with 5–15 mg 13-*cis*-RA/kg orally and examined implicit and explicit memory using a unique radial maze methodology. They describe deficits in explicit memory in rats treated with 10 or 15 mg/kg. Such deficits have been attributed to hippocampal alterations [12]. Again, the differences in testing methodology are substantial and it is unclear how implicit/explicit memory distinctions relate to working/reference memory differentiations.

The third maze used here, the NCTR complex maze, provides an assessment of spatial learning and memory which is not confounded by swim stress or hunger motivation. Similar to the effects using the Morris water and 8-arm radial mazes, 13-*cis*-RA treatment had no effects on learning or memory as measured by the NCTR complex maze. Rats treated with 13-*cis*-RA learned the appropriate path between the two goal arms, earning similar numbers of reinforcers as controls. The significant female advantage in complex maze performance has often been noted in this laboratory, although it has not always been significant. Here, however, likely due to the larger number of subjects, the effects were apparent statistically.

Morris water maze and complex maze endpoints measured here for the control group were very similar to those measured in control groups of previous studies [14,13]. In those two previous studies, one control group was untreated (i.e., no gavage or injection treatment) and housed individually [13] while the other control group was treated with saline on postnatal day 7 via subcutaneous injections and housed with same-sex siblings beginning at weaning [14]. This consistency in control group performance regardless of treatment or housing is reassuring with regard to methodology, particularly since it has been suggested that gavage stress might obscure effects of 13-*cis*-RA treatment [34]. However, such consistency should not indicate that these tests are insensitive to experimental treatments in our laboratory. For example, synthetic glucocorticoid treatment on PND 7 resulted in deficits in Morris water maze performance [14] and NCTR complex maze performance was quite sensitive to strain differences [13].

Performances in the complex maze and Morris water maze are dependent on spatial learning and memory; however, behavioral endpoints in these two mazes are not correlated with one another [15]. The two maze types are often differentially sensitive to various cognitive alterations (e.g., aging) [30] and some have suggested that appetitive and aversive spatial tasks are dissociable [11]. It was for these reasons that both maze types were included in the current study. Yet neither a water-motivated, food-motivated, or escape-motivated maze was sensitive to 13-*cis*-RA treatment.

In summary, chronic 13-*cis*-RA treatment had no effects on spatial learning and memory of male and female rats as measured by three maze tasks. Given the comparability of 13-*cis*-RA serum levels with the doses used here and those of humans prescribed Accutane®, the data are reassuring in that it seems unlikely that learning or memory deficits would result from 13-*cis*-RA treatment. If 13-*cis*-RA treatment does cause effects on learning, more challenging or demanding tasks may be necessary to detect subtle effects.

Acknowledgements

The authors would like to acknowledge the expert technical assistance of Mr. C. Delbert Law of the Bionetics Corporation. His skills and proficiencies surpass anything we might have hoped for in a technical specialist. Kimberly Berry was supported through a fellowship from the Oak Ridge Institute for Science and Education through an interagency agreement between the U.S. Department of Energy and the U.S. Food and Drug Administration.

References

[1] B. Almond-Roesler, U. Blume-Peytavi, S. Bisson, M. Krahn, E. Orfanos, C.E. Orfanos, Monitoring of isotretinoin therapy by measuring the plasma levels of isotretinoin and 4-oxo-isotretinoin. A useful tool for management of severe acne, Dermatology 196 (1998) 176–181.

[2] E. Baldi, M. Efoudebe, C.A. Lorenzini, C. Bucherelli, Spatial navigation in the Morris water maze: working and long lasting reference memories, Neurosci. Lett. 378 (2005) 176–180.

[3] J. Beiko, R. Lander, E. Hampson, F. Boon, D.P. Cain, Contribution of sex differences in the acute stress response to sex differences in water maze performance in the rat, Behav. Brain Res. 151 (2004) 239–253.

[4] R.K. Brazzell, F.M. Vane, C.W. Ehmann, W.A. Colburn, Pharmacokinetics of isotretinoin during repetitive dosing to patients, Eur. J. Clin. Pharmacol. 24 (1983) 695–702.

[5] M.Y. Chiang, D. Misner, G. Kempermann, T. Schikorski, V. Giguere, H.M. Sucov, F.H. Gage, C.F. Stevens, R.M. Evans, An essential role for retinoid receptors RARbeta and RXRgamma in long-term potentiation and depression, Neuron 21 (1998) 1353–1361.

[6] F.J. Cisneros, B.J. Gough, R.E. Patton, S.A. Ferguson, Serum levels of albumin, triglycerides, total protein and glucose in rats are altered after oral treatment with low doses of 13-*cis*-retinoic acid or all-*trans*-retinoic acid, J. Appl. Toxicol. 25 (2005) 470–478.

[7] S. Cocco, G. Diaz, R. Stancampiano, A. Diana, M. Carta, R. Curreli, L. Sarais, F. Fadda, Vitamin A deficiency produces spatial learning and memory impairment in rats, Neuroscience 115 (2002) 475–482.

[8] J. Crandall, Y. Sakai, J. Zhang, O. Koul, Y. Mineur, W.E. Crusio, P. McCaffery, 13-*cis*-retinoic acid suppresses hippocampal cell division and hippocampal-dependent learning in mice, Proc. Natl. Acad. Sci. U. S. A. 101 (2004) 5111–5116.

[9] R. D'Hooge, P.P. De Deyn, Applications of the Morris water maze in the study of learning and memory, Brain Res. Brain Res. Rev. 36 (2001) 60–90.

[10] M.M. Dopheide, R.E. Morgan, Isotretinoin (13-*cis*-retinoic acid) alters learning and memory, but not depression-like behaviors, in the adult rat, (submitted for publication).

[11] P.A. Dudchenko, J.P. Goodridge, D.A. Seiterle, J.S. Taube, Effects of repeated disorientation on the acquisition of spatial tasks in rats: dissociation between the appetitive radial arm maze and aversive water maze, J. Exp. Psychol., Anim. Behav. Processes 23 (1997) 194–210.

[12] N. Etchamendy, V. Enderlin, A. Marighetto, V. Pallet, P. Higueret, R. Jaffard, Vitamin A deficiency and relational memory deficit in adult mice: relationships with changes in brain retinoid signalling, Behav. Brain Res. 145 (2003) 37–49.

[13] S.A. Ferguson, A.M. Cada, Spatial learning/memory and social and nonsocial behaviors in the spontaneously hypertensive, Wistar-Kyoto and Sprague-Dawley rat strains, Pharmacol. Biochem. Behav. 77 (2004) 583–594.

[14] S.A. Ferguson, M.G. Paule, R.R. Holson, Neonatal dexamethasone on day 7 in rats causes behavioral alterations reflective of hippocampal, but not cerebellar, deficits, Neurotoxicol. Teratol. 23 (2001) 57–69.

[15] S.A. Ferguson, A. Pogge, T.A. Patterson, Relationships between spatial learning of a dry land maze and the Morris water maze in Sprague-Dawley (SD), Spontaneously Hypertensive (SHR), and Wistar-Kyoto (WKY) rat strains, Neurotoxicol. Teratol. 26 (2004) 504.

[16] S.A. Ferguson, F.J. Cisneros, B. Gough, J.P. Hanig, K.J. Berry, Chronic oral treatment with 13-*cis*-retinoic acid (isotretinoin) or all-*trans*-retinoic acid does not alter depression-like behaviors in rats, Toxicol. Sci. 87 (2005) 451–459.

[17] S.A. Ferguson, F.J. Cisneros, B.J. Gough, S.F. Ali, Four weeks of oral isotretinoin treatment causes few signs of general toxicity in male and female Sprague-Dawley rats, Food Chem. Toxicol. 43 (2005) 1289–1296.

[18] S.A. Ferguson, P.H. Siitonen, F.J. Cisneros, B. Gough, J.F. Young, Steady state pharmacokinetics of oral treatment with 13-*cis*-retinoic acid or all-*trans*-retinoic acid in male and female adult rats, Basic Clin. Pharmacol. Toxicol. 98 (2006) 582–587.

[19] S.A. Ferguson, F.J., Cisneros, J.P. Hanig, K.J. Berry, Chronic oral treatment with Accutane (13-*cis*-retinoic acid) does not increase measures of anhedonia or depression in male and female Sprague-Dawley rats, (in preparation).

[20] M. Gallagher, R. Burwell, M. Burchinal, Severity of spatial learning impairment in aging: development of a learning index for performance in the Morris water maze, Behav. Neurosci. 107 (1993) 618–626.

[21] R. Gerlai, Behavioral tests of hippocampal function: simple paradigms complex problems, Behav. Brain Res. 125 (2001) 269–277.

[22] C.W. Hendrix, K.A. Jackson, E. Whitmore, A. Guidos, R. Kretzer, C.M. Liss, L.P. Shah, K.C. Khoo, J. McLane, C.B. Trapnell, The effect of isotretinoin on the pharmacokinetics and pharmacodynamics of ethinyl estradiol and norethindrone, Clin. Pharmacol. Ther. 75 (2004) 464–475.

[23] I. Hoffman-La Roche, Accutane Product Information, (2006).

[24] R.R. Holson, S.F. Ali, A.C. Scallet, W. Slikker Jr., M.G. Paule, Benzodiazepine-like behavioral effects following withdrawal from chronic delta-9-tetrahydrocannabinol administration in rats, Neurotoxicology 10 (1989) 605–619.

[25] C.E. Hotchkiss, J.R. Latendresse, S.A. Ferguson, Oral treatment with 13-*cis*-and all-*trans*-retinoic acid decreases bone mass in rats. Comp. Med. (in press).

[26] Z. Jonasson, Meta-analysis of sex differences in rodent models of learning and memory: a review of behavioral and biological data, Neurosci. Biobehav. Rev. 28 (2005) 811–825.

[27] I.G. Kerr, M.E. Lippman, J. Jenkins, C.E. Myers, Pharmacology of 13-*cis*-retinoic acid in humans, Cancer Res. 42 (1982) 2069–2073.

[28] M.A. Lane, S.J. Bailey, Role of retinoid signalling in the adult brain, Prog. Neurobiol. 75 (2005) 275–293.

[29] P. McCaffery, J. Zhang, J.E. Crandall, Retinoic acid signaling and function in the adult hippocampus, J. Neurobiol. 66 (2006) 780–791.

[30] R.N. McLay, S.M. Freeman, R.E. Harlan, A.J. Kastin, J.E. Zadina, Tests used to assess the cognitive abilities of aged rats: their relation to each other

and to hippocampal morphology and neurotrophin expression, Gerontology 45 (1999) 143–155.

[31] D.L. Misner, S. Jacobs, Y. Shimizu, A.M. de Urquiza, L. Solomin, T. Perlmann, L.M. De Luca, C.F. Stevens, R.M. Evans, Vitamin A deprivation results in reversible loss of hippocampal long-term synaptic plasticity, Proc. Natl. Acad. Sci. U. S. A. 98 (2001) 11714–11719.

[32] I. Nulman, M. Berkovitch, J. Klein, A. Pastuszak, R.S. Lester, N. Shear, G. Koren, Steady-state pharmacokinetics of isotretinoin and its 4-oxo metabolite: implications for fetal safety, J. Clin. Pharmacol. 38 (1998) 926–930.

[33] J.L. Nunez, W.A. Koss, J.M. Juraska, Hippocampal anatomy and water maze performance are affected by neonatal cryoanesthesia in rats of both sexes, Horm. Behav. 37 (2000) 169–178.

[34] K.C. O'Reilly, J. Shumake, F. Gonzalez-Lima, M.A. Lane, S.J. Bailey, Chronic administration of 13-*cis*-retinoic acid increases depression-related behavior in mice, Neuropsychopharmacology 31 (2006) 1919–1927.

[35] C.E. Orfanos, C.C. Zouboulis, Oral retinoids in the treatment of seborrhoea and acne, Dermatology 196 (1998) 140–147.

[36] Y. Sakai, J.E. Crandall, J. Brodsky, P. McCaffery, 13-*cis* retinoic acid (Accutane) suppresses hippocampal cell survival in mice, Ann. N. Y. Acad. Sci. 1021 (2004) 436–440.

[37] R.S. Stern, Medication and medical service utilization for acne 1995-1998, J. Am. Acad. Dermatol. 43 (2000) 1042–1048.

[38] M. Wietrzych, H. Meziane, A. Sutter, N. Ghyselinck, P.F. Chapman, P. Chambon, W. Krezel, Working memory deficits in retinoid X receptor gamma-deficient mice, Learn. Mem. 12 (2005) 318–326.

[39] D.K. Wysowski, J. Swann, A. Vega, Use of isotretinoin (Accutane) in the United States: rapid increase from 1992 through 2000, J. Am. Acad. Dermatol. 46 (2002) 505–509.